17. The year 1900 AD is highly significant for geneticists due to (JIPMER 2013)
a) Discovery of genes
b) Principle of linkage
c) Chromosomal theory of heredity
d) **Rediscovery of Mendelism**

18. The phenotypic ratio of trihybrid cross in F_2 generation is (JIPMER 2016)
a) **27:9:9:9:3:3:3:1** b) 9:3:3:1
c) 1:4:6:4:1 d) 27:9:3:3:9:1:2:1

19. In a mutational event when adenine is replaced by guanine, it is the case of (AIPMT 2004)
a) Frameshift mutatin b) Transcription
c) **Transition** d) Transversion

20. Mutations can be induced with (AIPMT 2011)
a) **Gamma radiations** b) Infrared radiations
c) IAA d) Ethylene

21. The mechanism that causes a gene to move from one linkage group to another is called (AIPMT 2015, NEET (Phase – II) 2016)
a) **Translocation** b) Crossing over
c) Inversion d) Duplication

22. A point mutation comprising the substitution of a purine by pyrimidine is called (AIIMS 2002)
a) Transition b) Translocation
c) Deletion d) **Transversion**

23. Frameshift mutation occurs when (AIPMT 2008)
a) Base is substituted
b) **base is deleted or added**
c) Anticodons are absent
d) None of these

24. The distance between two genes in a chromosome is measured in cross-over units which represent (AIIMS 2008)
a) Ratio of crossing over between them
b) **Percentage of crossing over between them**
c) Number of crossing over between them
d) None of these

25. When a cluster of genes show linkage behaviour they (AIPMT 2003)
a) do not show a chromosome map
b) show recombination during meiosis
c) **do not show independent assortment**
d) induce cell division

26. Genetic map is one that (AIPMT 2003)
a) **Establish sites of the genes on a chromosome**
b) Establishes the various stages in gene evolution
c) Shows the stages during the cell division
d) Shows the distribution of various species in a region

27. After a mutation at a genetic locus of the character of an organism changes due to the change in (AIPMT 2004)
a) DNA replication
b) Protein synthesis pattern
c) RNA transcription pattern
d) **Protein structure**

28. In a hexaploidy wheat, the haploid (n) and basic (x) numbers of chromosomes are (AIPMT 2007)
a) **n =21 and x =7** b) n =7 and x =21
c) n =21 and x =21 d) n =21 and x =14

29. Point mutation involves (AIPMT 2009)
a) Deletion b) Insertion
c) **Change in single base pair**
d) duplication

30. Which one of the following is a wrong statement regarding mutations**?** (AIPMT 2012)
a) UV and Gamma rays are mutagens
b) **Change in a single base pair of DNA does not cause mutation**
c) Deletion and insertion of base pairs cause frame shift mutations.
d) Cancer cells commonly show chromosomal aberrations.

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